

"Inner product"

$$v = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_d \end{pmatrix} \quad u = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_d \end{pmatrix}$$

Inner product (generalization of the dot product) takes two vectors $v, u \in V$ and provides a scalar $\langle v|u \rangle$ that belongs to the field F .

ket: $|v\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_d \end{pmatrix}$
 bra: $\langle v| = (\alpha_1^* \dots \alpha_d^*)$

$$\langle v|u \rangle = v^\dagger u = (\alpha_1^* \dots \alpha_d^*) \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_d \end{pmatrix} = \sum_{i=1}^d \alpha_i^* \beta_i$$

$F = \mathbb{R}$



$$q\text{-norm} = \left(\sum_{i=1}^d |\alpha_i|^p \right)^{1/p}$$

Conjugate symmetry: $\langle v|u \rangle = \langle u|v \rangle^*$ $\Rightarrow v^\dagger u = u^\dagger v$

Length (norm) of a vector: $\|v\|_2 = \sqrt{\langle v|v \rangle} = \sqrt{\sum_{i=1}^d |\alpha_i|^2}$ (normalized vector $\|v\| = 1$)

Angle between two real vectors: $\cos \theta = \frac{\langle v|u \rangle}{\|v\| \|u\|}$ (orthogonal vectors $\langle v|u \rangle = 0$)

$\|v\| = \sqrt{5}$ and $\|u\| = \sqrt{10}$, l_2 norm

Unit vector: $\hat{v} = \frac{v}{\|v\|} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

